

Virtual Painkillers: Designing Accessible Virtual Reality Experiences for Helping Cancer Patients Manage Pain at Home

Maria Matsangidou*
CYENS Centre of Excellence
m.matsangidou@cyens.org.cy

Angelos P. Kassianos†
University of Cyprus
kassianos.angelos@ucy.ac.cy

Demetrios Papaioannou
CYENS Centre of Excellence &
University of Cyprus
dpapai03@ucy.ac.cy

Theodoros Solomou
CYENS Centre of Excellence &
University of Cyprus
tsolom01@cs.ucy.ac.cy

Maria Krini
Cyprus Association of Cancer
Patients and Friends
mariakr@pasykaf.org

Maria Karekla
University of Cyprus
mkarekla@ucy.ac.cy

Constantinos S. Pattichis
CYENS Centre of Excellence &
University of Cyprus
pattichi@ucy.ac.cy

ABSTRACT

More than half of cancer patients are dealing with moderately severe pain on a monthly basis and most of them report a breakthrough pain experience at least once. Despite the importance of pain management for cancer patients, cancer pain remains undertreated. With computer technology and especially Virtual Reality offering open endless opportunities for pain management, we must consider how low-cost home-based Virtual Reality for cancer patients can be sensitively designed to provide comfortable, enriching pain management experiences. Working closely with 51 cancer patients, medical and paramedical personnel, we co-designed an intelligent personalized mobile application to first collect ecologically momentary assessment data on symptoms like pain and fatigue and Health-Related Quality of Life and subsequently enhance symptom management of cancer patients at home. Through this paper, we thoroughly explain the screening process and quantitative analysis we run to identify which environments patients would like to receive as a Virtual Reality intervention that can facilitate the design of Virtual Reality interventions for cancer patients.

CCS CONCEPTS

• **Human-centered computing**; • **Human-computer interaction (HCI)**; • **Haptic devices**; • **User studies**;

*Share the first authorship

†Share the first authorship

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI '22 Extended Abstracts, April 29–May 05, 2022, New Orleans, LA, USA

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9156-6/22/04.

<https://doi.org/10.1145/3491101.3503562>

KEYWORDS

Virtual Reality, Pain Management, Breast Cancer Patients, Home-based Rehabilitation

ACM Reference Format:

Maria Matsangidou, Angelos P. Kassianos, Demetrios Papaioannou, Theodoros Solomou, Maria Krini, Maria Karekla, and Constantinos S. Pattichis. 2022. Virtual Painkillers: Designing Accessible Virtual Reality Experiences for Helping Cancer Patients Manage Pain at Home. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29–May 05, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3491101.3503562>

1 INTRODUCTION

The World Health Organization (WHO) identifies cancer as the second leading cause of mortality, accounting for 1 in 6 deaths in 2018, creating the urgent need for effective and accessible care for all patients [1]. In Europe, the Europe Beating Cancer Plan was set as a regional priority to improve prevention, treatment, and follow-up care [2]. One of the main objectives of the plan is the provision of a high quality of care, which includes the development of new technologies for personalized medicine. This is because of evidence that improvements in cancer diagnosis and treatment have resulted in more people living with cancer creating the need for personalized medicine [3]. In Europe 2020, the 5-year survival rate for cancer patients is up to 54.56% [4]. As a result, cancer survival rates are improving and therefore, patients' Health-Related Quality of Life (HRQoL) concerns policymakers and researchers worldwide as cancer patients, usually undergo painful medical processes including medical examinations, surgeries, biopsies, chemotherapies, and others, which cause distress, fear, concerns and affect their physical, cognitive, and mental well-being [5].

Two of the most severe symptoms that patients experience are pain and fatigue which severely impact their HRQoL. Particularly, the transition from diagnosis to primary treatment, and then, to life after treatment may cause significant pain and a reduction in the cancer patients' HRQoL, their mental health, response to treatment,

the treatment adherence, the disease progression, and potentially the disease recurrence [6]. This was documented in a meta-analysis of 330 studies that revealed an association between cancer-related distress and mortality rates [7]. At the same time, the most prevalent symptom patients experience is cancer-related fatigue with recommendations for treatment including behavioral, psychological and psychoeducational approaches [8]. Monitoring strategies to reduce fatigue are thus urgently needed [9]. In addition, patients need to learn how to develop self-regulated skills to improve their functioning, mental well-being and HRQoL [5].

Home-care provision is currently a major challenge; highlighted by European Commission's Joint Research Centre (JRC) which reported that in many European Union (EU) countries, cancer screenings and follow-ups are postponed, and treatments are delayed or modified due to risk of contagion from hospital visits during the Covid19 pandemic [8]. The goal, therefore, remains to maximize on the capacities afforded by new technologies to address this challenge through effective home-based care applications, accessible to all.

2 RELATED WORK

2.1 Pain and Cancer

In general, pain has been described as one of the most prevalent medical complaints of patients, yet clinicians find it difficult to treat because of its complicated and subjective nature [11] [12]. Pain is simultaneously classified as a physical and mental disturbing experience that may arise from a real or perceived injury [13]. As a result, the perception of pain can be both nociceptive and subjective, with the same sensory input of pain causing varying levels of intensity between different people and in different situations.

Previous research suggests that more than half of cancer patients are dealing with moderately severe pain on monthly basis and most of them report a breakthrough pain (i.e., unbearable pain that requires immediate release of morphine, hydromorphone, oxycodone, or oral transmucosal fentanyl citrate to be controlled) experience at least once within this timeframe [14]. For the treatment of cancer-related pain, around 80% of cancer patients are receiving prescription of analgesics and/or strong opioids, but around 60% of these patients, report inadequate pain relief which possess difficulties with their everyday life activities and overly reduce their HRQoL [14]. Patients in particular report that pain management is overlooked in their treatment, even though many studies suggest that managing pain properly can positively affect clinical outcomes, including symptom distress, HRQoL, satisfaction with care, and survival rates [14] [15] [16] [17].

As mentioned above, several pharmacological and non-pharmacological solutions are administered for the treatment of pain among cancer patients but, with a number of studies demonstrating substantial evidence that cancer pain is often inadequately managed [18] [19] [20] [21] [22] [23] [24] [25] [26]. The impact of insufficient treatment of pain is usually overwhelming affecting one out of three cancer patients [27], resulting in lessening the physical functioning, the psychological and emotional well-being, and the social interactions of the patient who is suffering from pain [28]. It is worth mentioning that a previous study acknowledges that the persistence of pain occurrence in cancer patients, may concern the

patient even more than the potential death and this worry aids the drive for physician-assisted suicide [29]. A systematic review of research identifies pain as a risk factor for suicide in patients with cancer and concludes that comprehensive cancer care should include pain management to minimize suicidal ideation [30].

Regardless of the importance of pain management in cancer, the pain prevalence has not changed significantly over the past decades [31]. As a result, cancer pain remains undertreated due to the lack of proper administration of pain assessment and subsequently management [27]. As of 2021, pain severity is mainly assessed through paper-based methods, which requires a significant amount of time from both sides (e.g., for the patient in pain to fill and for the professional to code and analyze). In addition, the assessment of Patient Reported Outcome Measures (PROMs) can complicate even more the pain assessment process due to the long timeframe between the patients' visits to the professionals. An example of the above is a patient who experiences severe pain multiple times per week. The patient has to wait to report this to the oncologist during their visit, since very few clinicians can offer telephone consultations due to their hectic schedules. By the time the patient visits the oncologist and reports about the symptoms they experienced over the previous weeks (or in some cases even months) they need to represent the pain retrospectively and cognitively, which often results in under- or over- estimation of pain resulting in imprecise evidence provided to the healthcare professional. As a result, the pain may not be accurately described, and so the patient may not receive the appropriate treatment [32]. Timely evaluations of pain intensity, worst pain occurrence and functionality impairments are considered of vast importance, as this can inform healthcare professionals on the pain patterns which can subsequently lead to adequate treatment [33]. Recent research demonstrates that being able to accurately assess pain, can positively reflect on the pain management practices selected by the professional, which will have a significant effect on pain reduction [34]. The goal therefore remains, to maximize on the capacities afforded by new technologies to address pain evaluation in real time and subsequently offer management options through effective home-based applications, accessible to all patients in need.

2.2 Ecological Momentary Assessment

The biases that occur when patients are called to retrospectively evaluate their symptoms like pain and fatigue, include recall bias and mood congruency. Recall bias refers to inaccuracies when any individual is called to cognitively represent health state outcomes that have occurred over the previous weeks or months [35]. Mood congruency refers to the use of personalized heuristics when reconstructing memories with patients recalling states that are more congruent with their present state of mood [36]. To this end, the increasing use of new technologies in healthcare have facilitated the use of modern methods to collect PROMs from patients including pain using Ecologically Momentary Assessment (EMA) approaches. EMA refers to the repeated collection of information about participants' real-time experiences in their natural environments [36]. The use of EMA approaches can help overcoming limitations of traditional retrospective assessment in healthcare which is time consuming, expensive, requires paper filling and coding and personnel for managing data [37] [38]. EMA can also provide a more robust

picture of patients' subjective experiences over time taking into advantage experiential variance. If used properly, EMA can help intervention developers to design and deliver targeted interventions that account for users' real-life experiences and their needs.

2.3 Pain and Virtual Reality

Over the past decades, a variety of non-pharmacological methods, such as distraction technique via imagery, meditation, relaxation, hypnosis, and positive thinking, have been used by clinicians to decrease the patients' perception of pain, with research to conclude that this can positively complement the treatment of pain, especially in cases of acute pain [39] [40] [41] [42] [43].

Computer technology has opened up endless opportunities for pain management [44][45] [46] [47]. Virtual Reality (VR) is an example of this type of technology since it allows patients to interact with a computer-simulated reality [48] and can function to either distract them from noticing incoming signals of pain [49], or change their relation to their pain (perspective taking hypothesis [50]) Even though, VR for the management of pain is based on the traditional distractive practices, however it seems to be a more advanced kind of analgesia, since it provides the patient with multisensory signals [51] [52]. Specifically, with the help of actions, distractive VR is able to alter the perception of pain [11]. This is because the attentional resources are limited; thus, the use of distraction is able to decrease the cognitive capacity the patient has to process painful signals.

Regarding the use of VR for pain management in cancer patients, the research is still in its infancy. Therefore, only few studies have examined the effectiveness of VR as a form of analgesia in cancer patients and also in combination with EMA data collection of pain. Specifically, most of the aforementioned studies, examined the effectiveness of VR during painful processes (e.g., chemotherapy), and reported promising results [53] [57] [58] [59]. Overall, as can be seen and by a systematic review on VR for pain management in cancer patients [56] not much has been done in the context of offering a home-based pain management care to help the cancer patient to independently manage pain from home. We believe that the current landscape in cancer patient healthcare together with limitations in current assessment practices for cancer-related pain create the need to propose a solution based on the principles of EMA and Ecologically Momentary Interventions (EMI).

Aims and Objectives: The aim is to follow a co-design approach with target users (cancer patients) and experts (healthcare professionals) to design a novel home-care intervention for cancer patients that will facilitate the collection of pain and other PROMs and deliver targeted interventions using an EMI approach. The objectives are to: (a) design the intelligent mobile system and (b) to select the EMI content with the most acceptable VR environments according to patients.

3 METHODS

3.1 Ethics

Participants were recruited from all national government hospitals that provide specialist care to individuals who are dealing with any type of cancer. Ethical approval was sought from the National bioethics committee (Reference Number: EEBK EP 2019.01.64). All participants signed a consent form prior to their recruitment.

3.2 Participants

Fifty-one cancer patients and medical and paramedical personnel involved to the study. Specifically, a total of 25 cancer patients from both genders (females = 21, 84%), with ages between 23 to 78 years ($M = 49.41$, $SD = 10.74$) and a total of 26 medical and paramedical personnel (females = 15, 65%), aged between 23 to 63 years ($M = 36.8$, $SD = 10.81$) participated. Most of the cancer patients were at the second stage of cancer ($n = 13$), while the rest were dealing with first ($n = 7$), and third ($n = 5$) stage. The majority of the patients were breast cancer patients ($n = 19$), and the rest were thyroid ($n = 2$), lymphoma, kidney, sarcoma, uterine ($n = 1$ respectively). Further to the cancer patients, the medical and paramedical personnel were mostly nursing staff ($n=18$), psychiatrists and psychologists ($n=5$), oncologists ($n=1$), managers and research directors ($n=2$).

3.3 Study Design and Procedure

The study design emerged from rigorous discussions with experts in the field of healthcare, specialized in cancer, social and computer scientists, and Human-Computer Interaction (HCI). To be informed by the bibliography a systematic review of research that examined the effectiveness of VR for pain management in cancer patients [53] was carried out. Data were collected over a months' period. Data were collected via questionnaires and focus groups.

3.4 The system

We designed an intelligent personalized mobile application to first collect EMA data on pain and HRQoL and subsequently enhance pain management of cancer patients. The application aims to offer to cancer patients an empowering space capable to: a) monitor the level of pain cancer patients encounter during the day; b) manage and reduce the pain via a personalized solution; c) help the cancer patient maintain emotional well-being on a daily basis.

Specifically, the application was installed on the patients' self-phones. Optionally a smart watch was linked with the application, this was done to continually monitor in real-time the heart rate of the patient. Once an elevated heart rate was detected (i.e., based on research elevated heart rate corresponds to an indication of pain[53] [54] [55]) the application suggested for the VR intervention to be held. Apart from the heart rate indicator, the application notified the patient three times per day to self-report on the levels of experiencing pain, fatigue, stress and HRQoL at the moment. The times they would receive the prompts to register their data, could be selected by patients themselves. Once the patient levels of pain were found to be aroused, the VR intervention was offered to the patient again. Validated cut-off points were used on each scale to identify elevated symptomatology: for example, pain. Figure 1 depicts a part of the application.

3.5 Apparatus

The interface of the VR application was developed using the Unity3D game engine¹. The VR environments were downloaded from YouTube² using the 4K Video Downloader³. YouTube videos were selected as they are available for no cost. One of the main

¹<https://www.unity.com/>

²<https://www.youtube.com/>

³<https://www.4kdownload.com/>

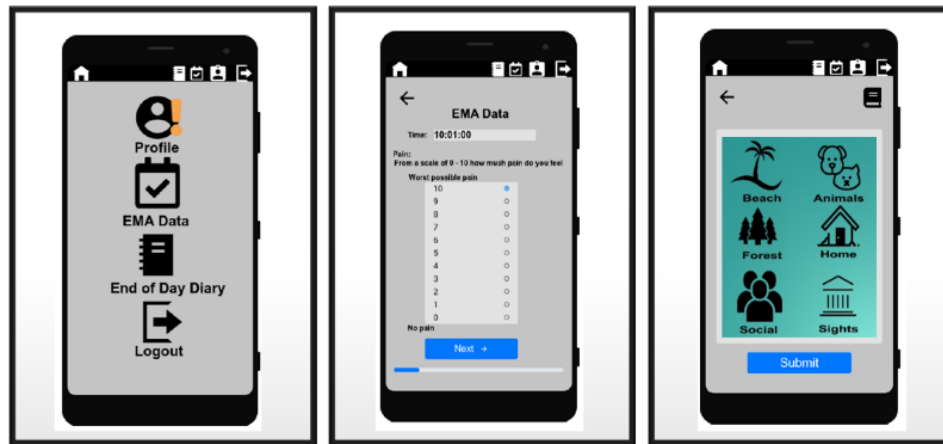


Figure 1: Depicts a part of the mobile application.

reasons of cancer-related distress is the added financial and emotional burden, therefore, we seek to design a low-cost affordable solution. The VR audial and visual content was steamed on any type of android self-phone. To convert the content into VR we used a low-cost Head-Mounted Display (HMD) (e.g., google cardboard⁴, Generisch VR Brille VR-Box VR-HMD⁵, to make sure that the system will be an affordable solution for all patients. The HMD is a wireless headset that can be used hands-free using its 3-point harness head strap and features optical lens with a 360° field of view. The combined weight of the HMD (480 grams) and phone (≈ 160 grams) is 640 grams, which makes it easy for the patient to use. Finally, a Garmin smartwatch⁶, was used to continually monitor in real-time the heart rate of the patient.

3.6 Materials

Questionnaires. Demographic characteristics of the participants were collected (i.e., sex, age, healthcare profession, type of cancer, stage of cancer, type of metastatic cancer if exist). Pain was measured on a scale range 0–10 during (0 = no pain, 10 = worst pain possible). Expectancy of managing pain via VR exposure was measured on a scale range 0–10 during (0 = strongly disagree, 10 = strongly agree). Content suggestion was measured through an open question (e.g., Let us know where you would like to be (give as much information as you can)).

Focus groups (n = 3). were organized at three national hospitals offering treatment to cancer patients. The focus groups were conducted to reflect on patients' and healthcare experts' preferences on VR environment content to complement palliative care therapy and define the aspects that must be avoided. Notes were taken to identify potential VR environments focused on the patients' interests and needs.

4 RESULTS

Three focus-groups with approximal duration of 60-minutes were conducted at three national hospitals offering treatment to cancer patients. Attendees were in total a group of 25 cancer patients and 26 specialists such as medical doctors, clinical psychologists, psychiatrists, nurses and managers within cancer healthcare sector. During the focus-group, the researchers presented the possibilities of VR technology and the results of the systematic literature review [56]. Afterwards, attendees were requested to brainstorm the type of VR content suitable for cancer patients and fill their choices into an online questionnaire. Attendees suggested the following categories: Animals/ Pets (kittens, puppies, safari); Arts Experience (cinema, concerts, museum); Blue and Green Nature (beach, forest, park); Familiar patient content / Personalized Content (family gatherings, home city); Hobbies and Sports (fishing, football); Home (garden, kitchen, living room); Social (pub, restaurant); and Travel (google maps, cities around the world).

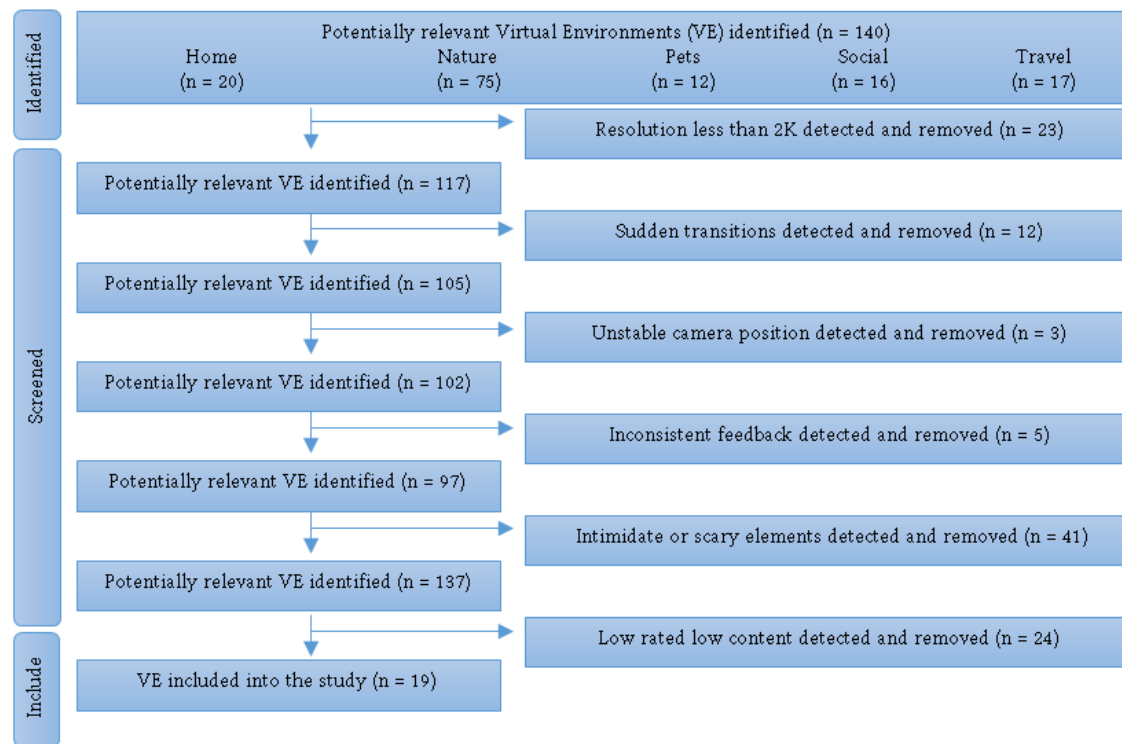
Once the above categories were recorded, we search systematically on YouTube for all the available 360° videos using the HCI Vargas-Avila and Hornbæk [60] methodology. We used the exact search terms aiming to cover any type of relevant Virtual Environments:

- **Animals / Pets:** Animas; Birds; Cats; Cow; Dogs; Ferrets; Fish; Gerbils; Goat; Guinea Pigs; Hamsters; Horses; Pets; Rabbits.
- **Arts Experience:** Art; Cinema; Concerts; Exhibits; Galleries; Museums; Sculptures.
- **Blue and Green Nature:** Beach; Flowers; Forests; Lakes; Mountains; Meadows; Nature; Ocean; Parks; Rivers; Sea; Trees; Valleys; Woodlands.
- **Familiar Patient-Content:** Birthday; Christmas; Celebration; Churches; Infant; Office; School; Temples; Villages.
- **Hobbies and Sports:** Basketball; Cooking; Fishing; Football; Hiking; Gardening; Sailing; Sport; Star(gazing); Tennis.
- **Home:** Apartment; Bedroom; Furniture; Garden; Home; House; Kitchen; Living Room; Porch.

⁴<https://arvr.google.com/cardboard/>

⁵<https://www.amazon.de/Generisch-Brille-VR-Box-Virtual-Reality-Wei%C3%9F/dp/B07ZNPH7ZW>

⁶<https://www.garmin.com/en-US/c/wearables-smartwatches/>



Flowchart 1: Presents the selection process of the VEs included into the system.

- **Social:** Bar; Café; Camp; Club; Festival; Holly Events; Library; Pub; Restaurant; Street Performance; Wedding; Christening.
- **Travel:** Austria; Belgium; Bulgaria; China; City; Cyprus; Czechia; Denmark; Estonia; Finland; France; Germany; Greece; Hungary; India; Island; Ireland; Italy; Japan; Latvia; Lithuania; Luxembourg; Malta; Netherlands; Poland; Portugal; Romania; Sights; Slovakia; Slovenia; Spain; Sweden; Travel; United Kingdom; United States⁷.

Overall, we identified 140 relevant virtual environments. Once the available virtual environments were identified based on the keywords and the participants interests, we imported them into a folder. Then, we excluded manually any content which was not consistent to the following inclusion criteria which were agreed by the research team. Exclusion criteria that were used to exclude the potential Virtual Environments were: 1) Resolution less than 2K (2048×1080) to avoid compromising resolution quality of content; 2) Sudden transitions between scenes to avoid being startled or confused; 3) Moving, shaking, unstable camera recording to avoid inducing motion sickness; 4) Animals or people that are close to the camera that may be perceived as startling or scary; 5) Negatively high arousing content that may be perceived as startling or scary; 6) Audial content that is not consistent with the visual content, aiming to provide coherent audial-visual feedback and avoid distraction; 7) Audial content that is perceived as loud, low, or noisy; 8) Explicit audial narration, as it is important for the cancer patients to hear the caregivers directions and prompts whilst using VR; 9) Computer generated content or special effects added onto 360o videos due to

the lack of evidence on the effects this type of content could have on patients.

Based on the above criteria, we identified 43 virtual environments within each category: Blue Nature (n=5)/ Green Nature (n=8); Home (n=8); Pets (n=6); Social (n=6); Travel (n=10). All participants were given pictures from the identified 43 virtual environments and were asked to rate positively their preferable environments. The results indicated that, the most attractive virtual environments where those which included blue-nature (nmax = 40) followed by green nature (nmax = 35) as well as those who presented home environments (nmax = 35), social events (e.g., Christmas) (nmax = 34), travelling (nmax = 29) and lastly pets (nmax = 26). We included in the study all the environment which received a rating over 50%. The highest rated videos were the blue-nature: forest beach (n = 40); sandy beach (n = 27); rocket beach(n = 26); green nature: snowy forest (n = 27); lake (n = 33); waterfalls (n = 35); woodland (n = 26); home environments: balcony (n = 35); gardens (n = 26); social events: Christmas market (n = 34), camp at Myrtos beach, Kefalonia – Greece (n = 26); street performance (n = 26); travelling: Big Ben (n = 26), UK; Delphy, Greece (n = 25); Hogia, Sophia (n = 25); and lastly animal / pets: elephants(n = 25); pinguins (n = 25); puppies (n = 26). The identification process is presented below (flowchart 1), and snapshots of the selected virtual environments are displayed in Figure 2

⁷Since the study run in the European Union, we restricted our search mostly in European Union Member States. We also added islands, India, Japan and United States, since they are common places to travel for our population.

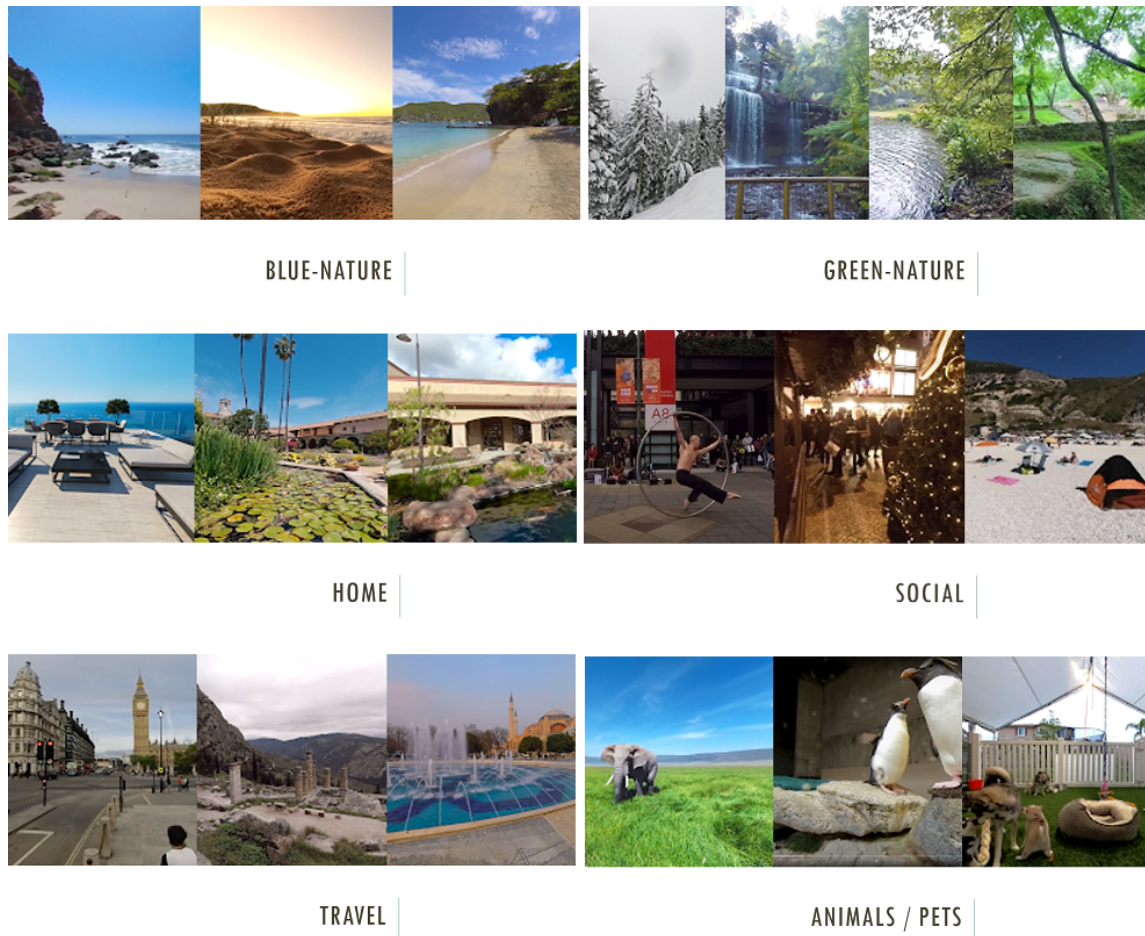


Figure 2: Depicts footages of the final VEs included into the system.

5 DISCUSSION AND FUTURE DIRECTIONS

Our results drawn from the three focus groups suggested eight main VR themes: (1) Animals/ Pets; (2) Arts Experience; (3) Blue and Green Nature; (4) Familiar patient content / Personalized Content; (5) Hobbies and Sports; (6) Home; (7) Social; and (8) Travel. Only five of the aforementioned themes were corresponding to existing 360° VR content (i.e., Blue and Green Nature; Home; Pets; Social; Travel). These themes indicated five main areas where cancer patients can be exposed to manage pain through VR experiences. In subsequent studies, we aim to test the application with cancer patients so to explicitly assess and explore the acceptability and effectiveness of these environments in real-time home settings to ameliorate the levels of pain, fatigue, stress and improve the patients' HRQoL.

It is also valuable to correlate the findings of this work with previous design research. Particularly, our survey with patients and medical and paramedical staff suggested for the system (solution) to incorporate:

Animals/ Pets. Previous research shows that having a pet can be a hope-engendering factor for the cancer patients, because the patient must direct his/her attention to the pet and take care of it. As a result, this has a positive impact on the cancer patient's health since

the patient it is forced to continue with everyday life activities [61]. It also suggests that pet ownership may be leverageable in interventions for mental health and wellbeing improvement during cancer recovery [62]. These are because pets are linked with positive emotions, meaningful lives, feeling of accomplishment and higher HRQoL [62] [63]. Interestingly, previous studies suggest that animals can enhance therapeutic healing with statistically significant health benefits and improvements in blood pressure, heart rate, salivary immunoglobulin A levels and offer reduction in depression and anxiety [64] [65] [66] [67] [67]. Lastly, previous research indicates that when patients view animal content on a screen this can have positive health benefits such as, the reduction in cardiovascular responses, stress, and anxiety [68].

Arts Experiences were also suggested as a content. Previous research recommends artistic content to enhance distraction and reduce perceived pain in clinical [69] [70] and non-clinical populations [72] [73]. Art was also applied in cancer patients for pain management demonstrating positive results [73].

As of *Nature*, the findings of previous studies advised that viewing nature can improve physiological (e.g., heart rate and blood pressure tend to decline within a few minutes of viewing nature) and

psychological (e.g., enhance emotional well-being, and aid recovery from stress) responses [74] [75]. Also, several studies demonstrated that specific environmental features can enhance distraction when experiencing acute pain and reduce specific types of pain (e.g., thermal pain can be reduced via viewing snowy environments) [76] [77] [78] [79]. Interestingly, our participants suggested that all natural environments should incorporate features which introduce cooling sensations (e.g., water elements / bluish features). Based on previous studies pain tolerance can be increased when a stimulus is linked to a blue visual cue [46] [80] [81].

Familiar patient content / Personalized Content and Home environment were also suggested. This is because many cancer patients might be living within hospital care units for care to be provided to them. Those patients usually wish to visit a memorable place or even to view their homes for one last time. Our findings are in line with a previous study which disclosed preliminary findings of terminal cancer patients who also suggested that they wish to visit a memorable place [82]. Our findings are also in line with studies which suggested that VR can be the solution for palliative care units with restrictions on visitations [83] [84]. Terminal patients who are at a grieving end of life stage reported that not being able to return at home or view their loved ones raised depression and sadness [85]. Therefore, VR can offer the possibility for “home-return” disregarding the hospital boundaries restrictions.

Hobbies and Sports were also reported as a preferable content by both patients and health care professionals. In line with previous research, it is suggested that caregivers believe that being able to offer to the terminal patient the ability to engage with preferable hobbies during end life period is empowering for both the patient and the therapist [86] [87]. In addition, past research suggested that the engagement with such activities can enhance the HRQoL for the patient in pain [88].

Social-interaction features were also suggested as previous research documented that the sense of belongingness can enhance the HRQoL for the cancer patients [89] [90] [91]. Particularly, it was found that access to social support and sense of belongingness can improve resilience coping mechanism in cancer patients which support the positive cognitive model of adjustment in emotionally distressing situations [92] [93]. Previous research also suggested that, having a cognitive model of adjustment based on spiritual resources, such as social support, can help cancer patients to cope with the emotional burden [94]. This was contrary to the common preconception that VR is an isolating experience [95]. Opposite to that notion, and in line with a study that examined the suitable types of VR for people with dementia [96], we found that cancer patients (similarly to dementia patients) suggested that VR can be a space where social interactions can be held. This was further supported by studies which suggested that VR can be the medium through which patients open up and communicate even more the feelings and ideas [97] [98]. Also, a study with patients suffering from body image concerns and eating disorders [98] which used a multi-user VR system where the therapist and the patient interacted on a remote basis, in a pleasant environment suggested that real-time communication can elicit via VR with therapeutic outcomes for the patient.

Finally, *Travel* was also suggested as a relevant theme. Based on previous research, terminal cancer patients are empowering by the

ability of traveling via Google Earth VR. This preliminary study suggested that VR travel can be efficacious and safe for terminal cancer patients for improving symptom related to pain, fatigue, shortness of breathing, depression, anxiety, and general well-being [82].

It is also valuable to reflect on the limitations of this work. Particularly, we found that a small number of oncologists participated to the study, this might be a signal of low uptake of intervention or promotion by oncologists. We also like to highlight the willingness of cancer patient groups with high survival rates (e.g., breast cancer) to participate to the study, as opposed to cancer patients' groups with lower survival rates and higher levels of pain (e.g., skin cancer patients).

To conclude, this paper presented a study with fifty-one cancer patients and medical and paramedical personnel focusing on the co-design of VR environments for cancer patients. Through iterative design process, we discuss the design of eight different VR themes. Through a systematic search we identified 140 different VR environments corresponding to the suggested themes. Based on the inclusion criteria and the second phase of the study (i.e., re-evaluation of the VR environments) only five out of eight themes were detected and included into the application. Finally, we assessed the selected VR environments and through a thorough screening process and quantitative analysis we identified which environments patients would like to receive as a VR intervention that can facilitate the design of VR interventions for this population, and we have selected the intervention content for the EMI which will be subsequently tested as to how this can be used together with an EMA approach using the system developed. A future study will assess the acceptability and effectiveness of this application.

ACKNOWLEDGMENTS

This work was co-funded by the European Regional Development Fund and the Republic of Cyprus through the Research and Innovation Foundation (project: OPPORTUNITY/0916/MSCA/0016). This project has also received funding from the European Union's Horizon 2020 Research and Innovation Programme under Grant Agreement No 739578 and the Government of the Republic of Cyprus through the Deputy Ministry of Research, Innovation and Digital Policy.

REFERENCES

- [1] World Health Organization. (n.d.). Cancer Overview Prevention and Management. World Health Organization. Retrieved October 14, 2021, from https://www.who.int/health-topics/cancer#tab=tab_1.
- [2] European Commission. (2020). ECIS - European Cancer Information System. ECIS. Retrieved October 6, 2021, from <https://ecis.jrc.ec.europa.eu/>.
- [3] Miller, K. D., Nogueira, L., Mariotto, A. B., Rowland, J. H., Yabroff, K. R., Alfano, C. M., ... & Siegel, R. L. (2019). Cancer treatment and survivorship statistics, 2019. CA: a cancer journal for clinicians, 69(5), 363-385.
- [4] European Federation of Pharmaceutical Industries and Associations (EFPIA). (n.d.). Cancer Care in 2020 – An overview of cancer outcomes data across Europe Survival. EFPIA Homepage. Retrieved October 14, 2021, from <http://www.efpia.eu/publications/cancer-comparator-report/survival>.
- [5] Armes, J., Crowe, M., Colbourne, L., Morgan, H., Murrells, T., Oakley, C., ... & Richardson, A. (2009). Patients' supportive care needs beyond the end of cancer treatment: a prospective, longitudinal survey. Journal of Clinical Oncology, 27(36), 6172-6179.
- [6] Zhang, Y., Zanos, P., Jackson, I. L., Zhang, X., Zhu, X., Gould, T., & Vujaskovic, Z. (2020). Psychological stress enhances tumor growth and diminishes radiation response in preclinical model of lung cancer. Radiotherapy and Oncology, 146, 126-135.

- [7] Chida, Y., Hamer, M., Wardle, J., & Steptoe, A. (2008). Do stress-related psychosocial factors contribute to cancer incidence and survival? *Nature clinical practice Oncology*, 5(8), 466-475.
- [8] Fabi, A., Bhargava, R., Fatigoni, S., Guglielmo, M., Horneber, M., Roila, F., ... & Ripamonti, C. I. (2020). Cancer-related fatigue: ESMO Clinical Practice Guidelines for diagnosis and treatment. *Annals of Oncology*, 31(6), 713-723.
- [9] Al Maqbali, M., Al Sinani, M., Al Naamani, Z., & Al Badi, K. (2021). Prevalence of fatigue in patients with cancer: a systematic review and meta-analysis. *Journal of Pain and Symptom Management*, 61(1), 167-189.
- [10] European Cancer. (n.d.). Impact of COVID-19 on Cancer. Impact of covid-19 on Cancer - European Cancer Organisation. Retrieved October 14, 2021, from <https://www.europecancercancer.org/2-standard/169-european-cancer-summit-2020-impact-covid-on-cancer>.
- [11] Gold, J. I., Belmont, K. A., & Thomas, D. A. (2007). The neurobiology of virtual reality pain attenuation. *CyberPsychology & Behavior*, 10(4), 536-544.
- [12] Mahrer, N. E., & Gold, J. I. (2009). The use of virtual reality for pain control: A review. *Current pain and headache reports*, 13(2), 100-109.
- [13] Merskey, H., & Bogduk, N. (1994). Classification of chronic pain, IASP Task Force on Taxonomy. Seattle, WA: International Association for the Study of Pain Press.
- [14] Breivik, H., Cherny, N., Collett, B., De Conno, F., Filbet, M., Foubert, A. J., ... & Dow, L. (2009). Cancer-related pain: a pan-European survey of prevalence, treatment, and patient attitudes. *Annals of oncology*, 20(8), 1420-1433.
- [15] Higginson, I. J., Finlay, I. G., Goodwin, D. M., Hood, K., Edwards, A. G., Cook, A., ... & Normand, C. E. (2003). Is there evidence that palliative care teams alter end-of-life experiences of patients and their caregivers? *Journal of pain and symptom management*, 25(2), 150-168.
- [16] Zimmermann, C., Riechelmann, R., Krzyzanowska, M., Rodin, G., & Tannock, I. (2008). Effectiveness of specialized palliative care: a systematic review. *Jama*, 299(14), 1698-1709.
- [17] Temel, J. S., Greer, J. A., Muzikansky, A., Gallagher, E. R., Admane, S., Jackson, V. A., ... & Lynch, T. J. (2010). Early palliative care for patients with metastatic non-small-cell lung cancer. *New England Journal of Medicine*, 363(8), 733-742.
- [18] Bruera, E., & Kim, H. N. (2003). Cancer pain. *Jama*, 290(18), 2476-2479.
- [19] Von Roenn, J. H., Cleeland, C. S., Gonin, R., Hatfield, A. K., & Pandya, K. J. (1993). Physician attitudes and practice in cancer pain management: a survey from the Eastern Cooperative Oncology Group. *Annals of Internal medicine*, 119(2), 121-126.
- [20] Cleeland, C. S., Gonin, R., Hatfield, A. K., Edmonson, J. H., Blum, R. H., Stewart, J. A., & Pandya, K. J. (1994). Pain and its treatment in outpatients with metastatic cancer. *New England Journal of Medicine*, 330(9), 592-596.
- [21] Cleeland, C. S. (1998). Undertreatment of cancer pain in elderly patients. *Jama*, 279(23), 1914-1915.
- [22] Pargson, K. L., & Hailey, B. J. (1999). Barriers to effective cancer pain management: a review of the literature. *Journal of pain and symptom management*, 18(5), 358-368.
- [23] Bernabei, R., Gambassi, G., Lapane, K., Landi, F., Gatsonis, C., Dunlop, R., ... & SAGE Study Group. (1998). Management of pain in elderly patients with cancer. *Jama*, 279(23), 1877-1882.
- [24] Larue, F., Colleau, S. M., Brasseur, L., & Cleeland, C. S. (1995). Multicentre study of cancer pain and its treatment in France. *Bmj*, 310(6986), 1034-1037.
- [25] MacDonald, N., Findlay, H. P., Bruera, E., Dudgeon, D., & Kramer, J. (1997). A Canadian survey of issues in cancer pain management. *Journal of pain and symptom management*, 14(6), 332-342.
- [26] Hsieh, R. K. (2005). Pain control in Taiwanese patients with cancer: a multicenter, patient-oriented survey. *Journal of the Formosan Medical Association*, 104(12), 913-919.
- [27] Greco, M. T., Roberto, A., Corli, O., Deandrea, S., Bandieri, E., Cavuto, S., & Apolone, G. (2014). Quality of cancer pain management: an update of a systematic review of undertreatment of patients with cancer. *J Clin Oncol*, 32(36), 4149-4154.
- [28] Serlin, R. C., Mendoza, T. R., Nakamura, Y., Edwards, K. R., & Cleeland, C. S. (1995). When is cancer pain mild, moderate or severe? Grading pain severity by its interference with function. *Pain*, 61(2), 277-284.
- [29] Nuland, S. B. (2000). Physician-assisted suicide and euthanasia in practice.
- [30] Anguiano, L., Mayer, D. K., Piven, M. L., & Rosenstein, D. (2012). A literature review of suicide in cancer patients. *Cancer nursing*, 35(4), 14-26.
- [31] Lohre, E. T., Thronæs, M., & Klepstad, P. (2020). Breakthrough cancer pain in 2020. *Current opinion in supportive and palliative care*, 14(2), 94-99.
- [32] Mirabile, A., Airolidi, M., Ripamonti, C., Bolner, A., Murphy, B., Russi, E., ... & Bossi, P. (2016). Pain management in head and neck cancer patients undergoing chemo-radiotherapy: Clinical practical recommendations. *Critical reviews in oncology/hematology*, 99, 100-106.
- [33] Fallon, M., Giusti, R., Aielli, F., Hoskin, P., Rolke, R., Sharma, M., & Ripamonti, C. I. (2018). Management of cancer pain in adult patients: ESMO Clinical Practice Guidelines. *Annals of Oncology*, 29, iv166-iv191.
- [34] Lohre, E. T., Thronæs, M., Brunelli, C., Kaasa, S., & Klepstad, P. (2020). An in-hospital clinical care pathway with integrated decision support for cancer pain management reduced pain intensity and needs for hospital stay. *Supportive Care in Cancer*, 28(2), 671-682.
- [35] Solhan, M. B., Trull, T. J., Jahng, S., & Wood, P. K. (2009). Clinical assessment of affective instability: comparing EMA indices, questionnaire reports, and retrospective recall. *Psychological assessment*, 21(3), 425.
- [36] Shiffman, S., Stone, A. A., & Hufford, M. R. (2008). Ecological momentary assessment. *Annu. Rev. Clin. Psychol.*, 4, 1-32.
- [37] Wright, E. P., Selby, P. J., Crawford, M., Gillibrand, A., Johnston, C., Perren, T. J., ... & Cull, A. (2003). Feasibility and compliance of automated measurement of quality of life in oncology practice. *Journal of Clinical Oncology*, 21(2), 374-382.
- [38] Mehl, M. R., & Holleran, S. E. (2007). An empirical analysis of the obtrusiveness of and participants' compliance with the electronically activated recorder (EAR). *European Journal of Psychological Assessment*, 23(4), 248-257.
- [39] Blount, R. L., Bachanas, P. J., Powers, S. W., Cotter, M. C., Franklin, A., Chaplin, W., Mayfield, J., Henderson, M., & Blount, S. D. (1992). Training children to cope and parents to coach them during routine immunizations: Effects on child, parent, and staff behaviors. *Behavior Therapy*, 23(4), 689-705.
- [40] Cohen, L. L., Blount, R. L., & Panopoulos, G. (1997). Nurse coaching and cartoon Distraction: An effective and practical intervention to reduce child, parent, and nurse distress during immunizations. *Journal of Pediatric Psychology*, 22(3), 355-370.
- [41] Jay, S., Elliott, C. H., Fitzgibbons, I., Woody, P., & Siegel, S. (1995). A comparative study of cognitive behavior therapy versus general anesthesia for painful medical procedures in children. *Pain*, 62(1), 3-9.
- [42] Linton, S. J. (1982). A critical review of behavioural treatments for chronic benign pain other than headache. *British Journal of Clinical Psychology*, 21(4), 321-337.
- [43] Patterson, D. R., & Ptacek, J. T. (1997). Baseline pain as a moderator of hypnotic analgesia for burn injury treatment. *Journal of Consulting and Clinical Psychology*, 65(1), 60-67.
- [44] Phelan, I., Arden, M., Matsangidou, M., Carrion-Plaza, A., & Lindley, S. (2021, May). Designing a Virtual Reality Myoelectric Prosthesis Training System for Amputees. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (pp. 1-7).
- [45] Phelan, I., Furness, P. J., Dunn, H. D., Carrion-Plaza, A., Matsangidou, M., Dimitri, P., & Lindley, S. (2021). Immersive virtual reality in children with upper limb injuries: Findings from a feasibility study. *Journal of Pediatric Rehabilitation Medicine*, (Preprint), 1-14.
- [46] Phelan, I., Furness, P. J., Matsangidou, M., Babiker, N. T., Fehily, O., Thompson, A., ... & Lindley, S. A. (2021). Designing effective virtual reality environments for pain management in burn-injured patients. *Virtual Reality*, 1-15.
- [47] Phelan, I., Furness, P. J., Matsangidou, M., Carrion-Plaza, A., Dunn, H., Dimitri, P., & Lindley, S. A. (2021). Playing your pain away: designing a virtual reality physical therapy for children with upper limb motor impairment. *Virtual Reality*, 1-13.
- [48] Brahmam, S., & Jain, L. C. (Eds.). (2011). *Advanced computational intelligence paradigms in healthcare 6: virtual reality in psychotherapy, rehabilitation, and assessment* (Vol. 337). Springer.
- [49] Matsangidou, M., Ang, C. S., & Sakel, M. (2017). Clinical utility of virtual reality in pain management: a comprehensive research review. *British Journal of Neuroscience Nursing*, 13(3), 133-143.
- [50] Karekla, M., Karademas, E., & Gloster, A. (2019). The Common Sense Model of self-regulation and Acceptance and Commitment Therapy: Integrating strategies to guide interventions for chronic illness. *Health Psychology Review*, 13, 490-503.
- [51] Hoffman, H. G., Seibel, E. J., Richards, T. L., Furness, T. A., Patterson, D. R., & Sharar, S. R. (2006). Virtual reality helmet display quality influences the magnitude of virtual reality analgesia. *The Journal of Pain*, 7(11), 843-850.
- [52] Schmitt, Y. S., Hoffman, H. G., Blough, D. K., Patterson, D. R., Jensen, M. P., Soltani, M., ... & Sharar, S. R. (2011). A randomized, controlled trial of immersive virtual reality analgesia, during physical therapy for pediatric burns. *Burns*, 37(1), 61-68.
- [53] Matsangidou, M., Ang, C. S., Mauger, A. R., Otkhmezuri, B., & Tabbaa, L. (2017, September). How real is unreal? In *IFIP Conference on Human-Computer Interaction* (pp. 273-288). Springer, Cham.
- [54] Matsangidou, M., Ang, C. S., Mauger, A. R., Intarasirisawat, J., Otkhmezuri, B., & Avraamides, M. N. (2019). Is your virtual self as sensational as your real? *Virtual Reality: The effect of body consciousness on the experience of exercise sensations. Psychology of sport and exercise*, 41, 218-224.
- [55] Matsangidou, M., Mauger, A. R., Ang, C. S., & Pattichis, C. S. (2020, July). Sampling electrocardiography conformation for a virtual reality pain management tool. In *International Conference on Human-Computer Interaction* (pp. 399-414). Springer, Cham.
- [56] Pittara, M., Matsangidou, M., Stylianides, K., Petkov, N., & Pattichis, C. S. (2020). *Virtual Reality for Pain Management in Cancer: A Comprehensive Review*. *IEEE Access*, 8, 225475-225489.
- [57] Schneider, S. M., Ellis, M., Coombs, W. T., Shonkwiler, E. L., & Folsom, L. C. (2003). Virtual reality intervention for older women with breast cancer. *CyberPsychology & Behavior*, 6(3), 301-307.
- [58] Wint, S. S., Eshelman, D., Steele, J., & Guzzetta, C. E. (2002, January). Effects of distraction using virtual reality glasses during lumbar punctures in adolescents with cancer. In *Oncology nursing forum* (Vol. 29, No. 1).

- [59] Gershon, J., Zimand, E., Lemos, R., Rothbaum, B. O., & Hodges, L. (2003). Use of virtual reality as a distractor for painful procedures in a patient with pediatric cancer: a case study. *CyberPsychology & Behavior*, 6(6), 657-661.
- [60] Bargas-Avila, J. A., & Hornbæk, K. (2011, May). Old wine in new bottles or novel challenges: a critical analysis of empirical studies of user experience. In *Proceedings of the SIGCHI conference on human factors in computing systems* (pp. 2689-2698).
- [61] Kylmä, J., & Juvakka, T. (2007). Hope in parents of adolescents with cancer—: Factors endangering and engendering parental hope. *European Journal of Oncology Nursing*, 11(3), 262-271.
- [62] Trigg, J. (2021). Examining the role of pets in cancer survivors' physical and mental wellbeing. *Journal of Psychosocial Oncology*, 1-20.
- [63] Chaudhary, S., & Srivastava, S. K. (2017). Own a Pet—Stay well and Happy: an exploratory study. *Indian Journal of Mental Health*, 4(4).
- [64] Allen, K., Shykoff, B. E., & Izzo Jr, J. L. (2001). Pet ownership, but not ACE inhibitor therapy, blunts home blood pressure responses to mental stress. *Hypertension*, 38(4), 815-820.
- [65] Luptak, J. E., & Nuzzo, N. A. (2004). The effects of small dogs on vital signs in elderly women: A pilot study. *Cardiopulmonary Physical Therapy Journal*, 15(1), 9.
- [66] Morrison, M. L. (2007). Health benefits of animal-assisted interventions. *Complementary health practice review*, 12(1), 51-62.
- [67] Stasi, M. F., Amati, D., Costa, C., Resta, D., Senepa, G., Scarafioiti, C., ... & Molaschi, M. (2004). Pet-therapy: a trial for institutionalized frail elderly patients. *Archives of gerontology and geriatrics. Supplement*, (9), 407-412.
- [68] Wells, D. L. (2005). The effect of videotapes of animals on cardiovascular responses to stress. *Stress and health*, 21(3), 209-213.
- [69] Shella, T. A. (2018). Art therapy improves mood, and reduces pain and anxiety when offered at bedside during acute hospital treatment. *The Arts in Psychotherapy*, 57, 59-64.
- [70] Trauger-Querry, B., & Haghighi, K. R. (1999). Balancing the focus: Art and music therapy for pain control and symptom management in hospice care. *The Hospice Journal*, 14(1), 25-38.
- [71] Koo, H. Y., Lim, J. H., & Park, H. R. (2007). Effects of distraction using balloon art on pain during intravenous injections in preschool children. *Child Health Nursing Research*, 13(1), 66-72.
- [72] Mitchell, L. A., MacDonald, R. A., & Knussen, C. (2008). An investigation of the effects of music and art on pain perception. *Psychology of Aesthetics, Creativity, and the Arts*, 2(3), 162.
- [73] Rhondali, W., Lasserre, E., & Filbet, M. (2012). Art therapy among palliative care inpatients with advanced cancer. *Palliative medicine*, 27(6), 571-572.
- [74] Wohlwill, J. F., & Weisman, G. D. (2012). *The physical environment and behavior: An annotated bibliography and guide to the literature*. Springer Science & Business Media.
- [75] Maller, C., Townsend, M., Pryor, A., Brown, P., & St Leger, L. (2006). Healthy nature healthy people: 'contact with nature' as an upstream health promotion intervention for populations. *Health promotion international*, 21(1), 45-54.
- [76] Carrougher, G. J., Hoffman, H. G., Nakamura, D., Lezotte, D., Soltani, M., Leahy, L., ... & Patterson, D. R. (2009). The effect of virtual reality on pain and range of motion in adults with burn injuries. *Journal of Burn Care & Research*, 30(5), 785-791.
- [77] Hoffman, H. G., Meyer III, W. J., Ramirez, M., Roberts, L., Seibel, E. J., Atzori, B., ... & Patterson, D. R. (2014). Feasibility of articulated arm mounted Oculus Rift Virtual Reality goggles for adjunctive pain control during occupational therapy in pediatric burn patients. *Cyberpsychology, Behavior, and Social Networking*, 17(6), 397-401.
- [78] Maani, C. V., Hoffman, H. G., Morrow, M., Maiers, A., Gaylord, K., McGhee, L. L., & DeSocio, P. A. (2011). Virtual reality pain control during burn wound debridement of combat-related burn injuries using robot-like arm mounted VR goggles. *The Journal of trauma*, 71(1 0), S125.
- [79] Schmitt, Y. S., Hoffman, H. G., Blough, D. K., Patterson, D. R., Jensen, M. P., Soltani, M., ... & Sharar, S. R. (2011). A randomized, controlled trial of immersive virtual reality analgesia, during physical therapy for pediatric burns. *Burns*, 37(1), 61-68.
- [80] Moseley, G. L., & Arntz, A. (2007). The context of a noxious stimulus affects the pain it evokes. *PAIN®*, 133(1-3), 64-71.
- [81] Martini, M., Pérez Marcos, D., & Sanchez-Vives, M. V. (2013). What color is my arm? Changes in skin color of an embodied virtual arm modulates pain threshold. *Frontiers in human neuroscience*, 7, 438-443.
- [82] Niki, K., Okamoto, Y., Maeda, I., Mori, I., Ishii, R., Matsuda, Y., ... & Uejima, E. (2019). A novel palliative care approach using virtual reality for improving various symptoms of terminal cancer patients: a preliminary prospective, multicenter study. *Journal of palliative medicine*, 22(6), 702-707.
- [83] Teo, W. Z., BcH, B. A. O., & Chai, Y. W. *Virtual Reality as a Bridge in Palliative Care during COVID-19*.
- [84] Niki, K., Okamoto, Y., & Ueda, M. (2020). Response to Wang *et al.*, Virtual Reality as a Bridge in Palliative Care during COVID-19 (DOI: 10.1089/jpm. 2020.0212). *Journal of palliative medicine*, 23(7), 892-894.
- [85] Otani, H., Yoshida, S., Morita, T., Aoyama, M., Kizawa, Y., Shima, Y., ... & Miyashita, M. (2017). Meaningful communication before death, but not present at the time of death itself, is associated with better outcomes on measures of depression and complicated grief among bereaved family members of cancer patients. *Journal of pain and symptom management*, 54(3), 273-279.
- [86] Claxton-Oldfield, S., MacDonald, J., & Claxton-Oldfield, J. (2006). What palliative care volunteers would like to know about the patients they are being asked to support. *American Journal of Hospice and Palliative Medicine®*, 23(3), 192-196.
- [87] Keesing, S., & Rosenwax, L. (2011). Is occupation missing from occupational therapy in palliative care? *Australian Occupational Therapy Journal*, 58(5), 329-336.
- [88] Chui, Y. Y., Kuan, H. Y., Fu, I. C., Liu, R. K., Sham, M. K., & Lau, K. S. (2009). Factors associated with lower quality of life among patients receiving palliative care. *Journal of advanced nursing*, 65(9), 1860-1871.
- [89] Ell, K., Nishimoto, R., Mediansky, L., Mantell, J., & Hamovitch, M. (1992). Social relations, social support and survival among patients with cancer. *Journal of psychosomatic research*, 36(6), 531-541.
- [90] Spiegel, D., Butler, L. D., Giese-Davis, J., Koopman, C., Miller, E., DiMiceli, S., ... & Kraemer, H. C. (2007). Effects of supportive-expressive group therapy on survival of patients with metastatic breast cancer: a randomized prospective trial. *Cancer*, 110(5), 1130-1138.
- [91] Wellisch, D. K., Mosher, M. B., & Van Scoy, C. (1978). Management of family emotion stress: Family group therapy in a private oncology practice. *International Journal of Group Psychotherapy*, 28(2), 225-231.
- [92] Alizadeh, S., Khanahmadi, S., Vedadhir, A., & Barjasteh, S. (2018). The relationship between resilience with self-compassion, social support and sense of belonging in women with breast cancer. *Asian Pacific journal of cancer prevention: APJCP*, 19(9), 2469-2474.
- [93] Helgeson, V. S., & Cohen, S. (1999). Social support and adjustment to cancer: reconciling descriptive, correlational, and intervention research. *Health Psychology*, 15(2): 135-148.
- [94] Lynn Gall, T., & Cornblat, M. W. (2002). Breast cancer survivors give voice: a qualitative analysis of spiritual factors in long-term adjustment. *Psycho-Oncology: Journal of the Psychological, Social and Behavioral Dimensions of Cancer*, 11(6), 524-535.
- [95] Pringle, R. (2017). Virtual reality is still too isolating to be 'the next big thing' in tech. *CBC News*, 14.
- [96] Hodge, J., Balaam, M., Hastings, S., & Morrissey, K. (2018, April). Exploring the design of tailored virtual reality experiences for people with dementia. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (pp. 1-13).
- [97] Blythe, M., Wright, P., Bowers, J., Boucher, A., Jarvis, N., Reynolds, P., & Gaver, B. (2010, August). Age and experience: ludic engagement in a residential care setting. In *Proceedings of the 8th ACM Conference on Designing Interactive Systems* (pp. 161-170).
- [98] Matsangidou, M., Otkhmezuri, B., Ang, C. S., Avraamides, M., Riva, G., Gaggioli, A., ... & Karekla, M. (2020). "Now i can see me" designing a multi-user virtual reality remote psychotherapy for body weight and shape concerns. *Human-Computer Interaction*, 1-27.